Breaking the Communication-Accuracy Trade-off: A Sparsified Information Diffusion Framework for Multi-Agent Collaborative Perception

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Appendix A
CKF Algorithm

The traditional Cubature Kalman Filter (CKF) algorithm [1] is illustrated in Alg. 1.

Appendix B
Derivation of CIF

First, A lemma to be used in the derivation is stated.

Lemma 1 ([2]). *For non-singular matrices $\boldsymbol{A} \in \mathbb{R}^{n\times n}$, $\boldsymbol{B} \in \mathbb{R}^{n\times m}$, $\boldsymbol{C} \in \mathbb{R}^{m\times n}$, and $\boldsymbol{D} \in \mathbb{R}^{m\times m}$, the following matrix inversion equality holds*

$$(\boldsymbol{A}-\boldsymbol{B}\boldsymbol{D}^{-1}\boldsymbol{C})^{-1} = \boldsymbol{A}^{-1}+\boldsymbol{A}^{-1}\boldsymbol{B}(\boldsymbol{D}-\boldsymbol{C}\boldsymbol{A}^{-1}\boldsymbol{B})^{-1}\boldsymbol{C}\boldsymbol{A}^{-1}. \tag{1}$$

Next, we derive the CIF algorithm on the basis of the CKF.

From the update of the error covariance matrix in CKF, it can be derived that the information matrix follows

$$\boldsymbol{Y}_k^i = (\boldsymbol{P}_k^i)^{-1} = \left[\boldsymbol{P}_{k|k-1}^i - \boldsymbol{K}_k^i \boldsymbol{P}_{x_k,z_k}^i (\boldsymbol{K}_k^i)^{\mathrm{T}}\right]^{-1}, \tag{2}$$

where

$$\boldsymbol{K}_k^i = \boldsymbol{P}_{x_k,z_k}^i (\boldsymbol{P}_{z_k,z_k}^i)^{-1}. \tag{3}$$

Eq. (2) can be further written as

$$\boldsymbol{Y}_k^i = \left[\boldsymbol{P}_{k|k-1}^i - \boldsymbol{P}_{x_k,z_k}^i (\boldsymbol{P}_{z_k,z_k}^i)^{-1} (\boldsymbol{P}_{x_k,z_k}^i)^{\mathrm{T}}\right]^{-1}. \tag{4}$$

According to Lemma 1, we take $\boldsymbol{A} = \boldsymbol{P}_{k|k-1}^i$, $\boldsymbol{B} = \boldsymbol{P}_{x_k,z_k}^i$, $\boldsymbol{C} = (\boldsymbol{P}_{x_k,z_k}^i)^{\mathrm{T}}$, and $\boldsymbol{D} = \boldsymbol{P}_{z_k,z_k}^i$. Then, Eq. (4) is equivalent to

$$\begin{aligned}(\boldsymbol{P}_k^i)^{-1} =& (\boldsymbol{P}_{k|k-1}^i)^{-1} + (\boldsymbol{P}_{k|k-1}^i)^{-1} \boldsymbol{P}_{x_k,z_k}^i \\ &\times \left[\boldsymbol{P}_{z_k,z_k}^i - (\boldsymbol{P}_{x_k,z_k}^i)^{\mathrm{T}} (\boldsymbol{P}_{k|k-1}^i)^{-1} \boldsymbol{P}_{x_k,z_k}^i\right]^{-1} \\ &\times (\boldsymbol{P}_{x_k,z_k}^i)^{\mathrm{T}} (\boldsymbol{P}_{k|k-1}^i)^{-1}.\end{aligned} \tag{5}$$

Applying the pseudo measurement matrix defined as

$$\boldsymbol{H}_k^i \triangleq (\boldsymbol{P}_{x_k,z_k}^i)^{\mathrm{T}} (\boldsymbol{P}_{k|k-1}^i)^{-1} = (\boldsymbol{P}_{x_k,z_k}^i)^{\mathrm{T}} \boldsymbol{Y}_{k|k-1}^i, \tag{6}$$

one has $\boldsymbol{P}_{x_k,z_k}^i = \boldsymbol{P}_{k|k-1}^i (\boldsymbol{H}_k^i)^{\mathrm{T}}$, which can be taken into (5) and results in

$$(\boldsymbol{P}_k^i)^{-1} = (\boldsymbol{P}_{k|k-1}^i)^{-1} + (\boldsymbol{H}_k^i)^{\mathrm{T}} (\boldsymbol{P}_{z_k,z_k}^i - \boldsymbol{\mathcal{P}}_k^i)^{-1} \boldsymbol{H}_k^i, \tag{7}$$

Algorithm 1: Cubature Kalman Filter (CKF) [1]

Input: initial target state $\boldsymbol{x}_0$, measurement $\boldsymbol{z}_k^i$.
Output: state estimate $\hat{\boldsymbol{x}}_k^i$ and error covariance matrix $\boldsymbol{P}_k^i$ at each time instant k.

Initialization:
$\hat{\boldsymbol{x}}_0^i = \mathbb{E}\{\boldsymbol{x}_0\},\ \boldsymbol{P}_0^i = \mathbb{E}\{(\boldsymbol{x}_0 - \hat{\boldsymbol{x}}_0^i)(\boldsymbol{x}_0 - \hat{\boldsymbol{x}}_0^i)^{\mathrm{T}}\}$.
for $k \leftarrow 1$ to T **do**
  Phase 1: Prediction
  // Generate $2n$ cubature points. $r = 1, \ldots, 2n$.
  $\boldsymbol{x}_{k-1}^{i,r} = \sqrt{\boldsymbol{P}_{k-1}^i}\boldsymbol{c}_r + \hat{\boldsymbol{x}}_{k-1}^i$,
  $\boldsymbol{c}_r = \begin{cases}\sqrt{n}\boldsymbol{e}_r, & 1 \le r \le n, \\ -\sqrt{n}\boldsymbol{e}_{r-n}, & n+1 \le r \le 2n,\end{cases}$
  $\boldsymbol{e}_r = [\underbrace{0,\ldots,0}_{r-1}, 1, \underbrace{0,\ldots,0}_{n-r}]^{\mathrm{T}}, 1 \le r \le n.$
  // Calculate the prior estimate and covariance.
  $\boldsymbol{x}_{k|k-1}^{i,r} = f(\boldsymbol{x}_{k-1}^{i,r})$,
  $\hat{\boldsymbol{x}}_{k|k-1}^i = \frac{1}{2n}\sum_{r=1}^{2n} \boldsymbol{x}_{k|k-1}^{i,r}$,
  $\boldsymbol{P}_{k|k-1}^i = \frac{1}{2n}\sum_{r=1}^{2n} \boldsymbol{x}_{k|k-1}^{i,r}(\boldsymbol{x}_{k|k-1}^{i,r})^{\mathrm{T}} - \hat{\boldsymbol{x}}_{k|k-1}^i(\hat{\boldsymbol{x}}_{k|k-1}^i)^{\mathrm{T}} + \boldsymbol{Q}_{k-1}$.
  Phase 2: Update
  // Obtain the predicted measurement.
  $\boldsymbol{x}_{k|k-1}^{i,r} = \sqrt{\boldsymbol{P}_{k|k-1}^i}\boldsymbol{c}_r + \hat{\boldsymbol{x}}_{k|k-1}^i$,
  $\boldsymbol{z}_{k|k-1}^{i,r} = h(\boldsymbol{x}_{k|k-1}^{i,r})$,
  $\hat{\boldsymbol{z}}_{k|k-1}^i = \frac{1}{2n}\sum_{r=1}^{2n} \boldsymbol{z}_{k|k-1}^{i,r}$.
  // Determine the measurement covariance.
  $\boldsymbol{P}_{z_k,z_k}^i = \frac{1}{2n}\sum_{r=1}^{2n} \boldsymbol{z}_{k|k-1}^{i,r}(\boldsymbol{z}_{k|k-1}^{i,r})^{\mathrm{T}} - \hat{\boldsymbol{z}}_{k|k-1}^i(\hat{\boldsymbol{z}}_{k|k-1}^i)^{\mathrm{T}} + \boldsymbol{R}_k^i$,
  $\boldsymbol{P}_{x_k,z_k}^i = \frac{1}{2n}\sum_{r=1}^{2n} \boldsymbol{x}_{k|k-1}^{i,r}(\boldsymbol{z}_{k|k-1}^{i,r})^{\mathrm{T}} - \hat{\boldsymbol{x}}_{k|k-1}^i(\hat{\boldsymbol{z}}_{k|k-1}^i)^{\mathrm{T}}$.
  // Compute the filter gain.
  $\boldsymbol{K}_k^i = \boldsymbol{P}_{x_k,z_k}^i(\boldsymbol{P}_{z_k,z_k}^i)^{-1}$.
  // Obtain the posterior estimate and covariance.
  $\hat{\boldsymbol{x}}_k^i = \hat{\boldsymbol{x}}_{k|k-1}^i + \boldsymbol{K}_k^i(\boldsymbol{z}_k^i - \hat{\boldsymbol{z}}_{k|k-1}^i)$,
  $\boldsymbol{P}_k^i = \boldsymbol{P}_{k|k-1}^i - \boldsymbol{K}_k^i \boldsymbol{P}_{z_k,z_k}^i (\boldsymbol{K}_k^i)^{\mathrm{T}}$.
end

where $\boldsymbol{\mathcal{P}}_k^i = \boldsymbol{H}_k^i \boldsymbol{P}_{k|k-1}^i (\boldsymbol{H}_k^i)^{\mathrm{T}}$. Eq. (7) can be written as

$$\boldsymbol{Y}_k^i = \boldsymbol{Y}_{k|k-1}^i + \underbrace{(\boldsymbol{H}_k^i)^{\mathrm{T}} (\boldsymbol{P}_{z_k,z_k}^i - \boldsymbol{\mathcal{P}}_k^i)^{-1} \boldsymbol{H}_k^i}_{\boldsymbol{I}_k^i}. \tag{8}$$

From (8) and (6), one can derive the correlation information $\boldsymbol{I}_k^i$ as

$$\boldsymbol{I}_k^i = \boldsymbol{Y}_{k|k-1}^i \boldsymbol{P}_{x_k,z_k}^i (\boldsymbol{D}_k^i)^{-1} (\boldsymbol{P}_{x_k,z_k}^i)^{\mathrm{T}} \boldsymbol{Y}_{k|k-1}^i, \tag{9}$$

where

$$\begin{aligned} \boldsymbol{D}_k^i =& \boldsymbol{P}_{z_k,z_k}^i - \boldsymbol{\mathcal{P}}_k^i \\ =& \boldsymbol{P}_{z_k,z_k}^i - \boldsymbol{H}_k^i \boldsymbol{P}_{k|k-1}^i (\boldsymbol{H}_k^i)^{\mathrm{T}} \\ =& \boldsymbol{P}_{z_k,z_k}^i - (\boldsymbol{P}_{x_k,z_k}^i)^{\mathrm{T}} \boldsymbol{Y}_{k|k-1}^i \boldsymbol{P}_{x_k,z_k}^i. \end{aligned} \tag{10}$$

Further writing the filter gain as

$$\boldsymbol{K}_k^i = \boldsymbol{P}_k^i (\boldsymbol{P}_k^i)^{-1} \boldsymbol{P}_{k|k-1}^i (\boldsymbol{H}_k^i)^{\mathrm{T}} (\boldsymbol{P}_{z_k,z_k}^i)^{-1}, \tag{11}$$

and substituting (7) into (11), one has

$$\begin{aligned} \boldsymbol{K}_k^i =& \boldsymbol{P}_k^i [(\boldsymbol{H}_k^i)^{\mathrm{T}} + (\boldsymbol{H}_k^i)^{\mathrm{T}} (\boldsymbol{P}_{z_k,z_k}^i - \boldsymbol{\mathcal{P}}_k^i)^{-1} \boldsymbol{\mathcal{P}}_k^i] (\boldsymbol{P}_{z_k,z_k}^i)^{-1} \\ =& \boldsymbol{P}_k^i (\boldsymbol{H}_k^i)^{\mathrm{T}} (\boldsymbol{P}_{z_k,z_k}^i - \boldsymbol{\mathcal{P}}_k^i)^{-1}. \end{aligned} \tag{12}$$

Based on (12), rewrite the estimate's update equation as

$$\hat{\boldsymbol{x}}_k^i = \hat{\boldsymbol{x}}_{k|k-1}^i + \boldsymbol{P}_k^i (\boldsymbol{H}_k^i)^{\mathrm{T}} (\boldsymbol{P}_{z_k,z_k}^i - \boldsymbol{\mathcal{P}}_k^i)^{-1} \Delta \boldsymbol{z}_k^i, \tag{13}$$

where $\Delta \boldsymbol{z}_k^i = \boldsymbol{z}_k^i - \hat{\boldsymbol{z}}_{k|k-1}^i$. Considering (7), it can be further obtained that

$$\begin{aligned} \hat{\boldsymbol{y}}_k^i =& (\boldsymbol{P}_k^i)^{-1} \hat{\boldsymbol{x}}_k^i \\ =& (\boldsymbol{P}_k^i)^{-1} \hat{\boldsymbol{x}}_{k|k-1}^i + (\boldsymbol{H}_k^i)^{\mathrm{T}} (\boldsymbol{P}_{z_k,z_k}^i - \boldsymbol{\mathcal{P}}_k^i)^{-1} \Delta \boldsymbol{z}_k^i \\ =& (\boldsymbol{P}_{k|k-1}^i)^{-1} \hat{\boldsymbol{x}}_{k|k-1}^i + (\boldsymbol{H}_k^i)^{\mathrm{T}} (\boldsymbol{P}_{z_k,z_k}^i - \boldsymbol{\mathcal{P}}_k^i)^{-1} \\ & \times (\Delta \boldsymbol{z}_k^i + \boldsymbol{H}_k^i \hat{\boldsymbol{x}}_{k|k-1}^i) \\ =& \hat{\boldsymbol{y}}_{k|k-1}^i + \underbrace{(\boldsymbol{H}_k^i)^{\mathrm{T}} (\boldsymbol{P}_{z_k,z_k}^i - \boldsymbol{\mathcal{P}}_k^i)^{-1} (\Delta \boldsymbol{z}_k^i + \boldsymbol{H}_k^i \hat{\boldsymbol{x}}_{k|k-1}^i)}_{\boldsymbol{i}_k^i}, \end{aligned} \tag{14}$$

from which, one derives the information contribution $\boldsymbol{i}_k^i$ as

$$\begin{aligned} \boldsymbol{i}_k^i =& \boldsymbol{Y}_{k|k-1}^i \boldsymbol{P}_{x_k,z_k}^i (\boldsymbol{D}_k^i)^{-1} \\ & \times \left[\Delta \boldsymbol{z}_k^i + (\boldsymbol{P}_{x_k,z_k}^i)^{\mathrm{T}} \boldsymbol{Y}_{k|k-1}^i \hat{\boldsymbol{x}}_{k|k-1}^i \right]. \end{aligned} \tag{15}$$

Appendix C
Derivation of ET-CKF

A lemma is introduced first before the derivation.

Lemma 2 ([3]). *For any given $\boldsymbol{x}, \boldsymbol{y} \in \mathbb{R}^n$ and scalar $\sigma > 0$, it follows*

$$\boldsymbol{x}\boldsymbol{y}^{\mathrm{T}} + \boldsymbol{y}\boldsymbol{x}^{\mathrm{T}} \leq \sigma \boldsymbol{x}\boldsymbol{x}^{\mathrm{T}} + \sigma^{-1} \boldsymbol{y}\boldsymbol{y}^{\mathrm{T}}. \tag{16}$$

Next, we derive the ET-CKF algorithm.

For node ν_i at time instant k, define the prediction error $\tilde{\boldsymbol{x}}_{k|k-1}^i$ and estimation error $\tilde{\boldsymbol{x}}_k^i$ as

$$\begin{cases} \tilde{\boldsymbol{x}}_{k|k-1}^i = \boldsymbol{x}_k - \hat{\boldsymbol{x}}_{k|k-1}^i, \\ \tilde{\boldsymbol{x}}_k^i = \boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^i, \end{cases} \tag{17}$$

respectively, where $\hat{\boldsymbol{x}}_{k|k-1}^i$ stands for the prior estimate, and $\hat{\boldsymbol{x}}_k^i$ represents the posterior estimate. Recall the posterior estimate of ET-CKF as

$$\hat{\boldsymbol{x}}_k^i = \hat{\boldsymbol{x}}_{k|k-1}^i + \bar{\boldsymbol{K}}_k^i (\boldsymbol{z}_{\tau_k^i}^i - \hat{\boldsymbol{z}}_{k|k-1}^i), \tag{18}$$

where $\bar{\boldsymbol{K}}_k^i = \gamma_k^i \boldsymbol{K}_k^i + (1 - \gamma_k^i) \boldsymbol{M}_k^i$ is the updated filter gain. In cases where the measurement is not triggered, we aim to derive the optimal filter gain $\boldsymbol{M}_k^i$. Then, Eq. (18) under $\gamma_k^i = 0$ follows

$$\hat{\boldsymbol{x}}_k^i = \hat{\boldsymbol{x}}_{k|k-1}^i + \boldsymbol{M}_k^i (\boldsymbol{z}_{\tau_{k-1}^i}^i - \boldsymbol{z}_k^i) + \boldsymbol{M}_k^i (\boldsymbol{z}_k^i - \hat{\boldsymbol{z}}_{k|k-1}^i). \tag{19}$$

Substituting (19) into (17) yields

$$\tilde{\boldsymbol{x}}_k^i = \tilde{\boldsymbol{x}}_{k|k-1}^i - \boldsymbol{M}_k^i \boldsymbol{\kappa}_k^i - \boldsymbol{M}_k^i (\boldsymbol{z}_k^i - \hat{\boldsymbol{z}}_{k|k-1}^i), \tag{20}$$

where $\boldsymbol{\kappa}_k^i = \boldsymbol{z}_{\tau_{k-1}^i}^i - \boldsymbol{z}_k^i$, and $\hat{\boldsymbol{z}}_{k|k-1}^i = h^i(\hat{\boldsymbol{x}}_{k|k-1}^i)$. From the Taylor expansion, the continuously differentiable measurement function $h^i(\cdot)$ at $\boldsymbol{x}_k$ satisfies

$$h^i(\boldsymbol{x}_k) = h^i(\hat{\boldsymbol{x}}_{k|k-1}^i) + \boldsymbol{G}_k^i \tilde{\boldsymbol{x}}_{k|k-1}^i + \phi^i(\boldsymbol{x}_k, \hat{\boldsymbol{x}}_{k|k-1}^i), \tag{21}$$

where $\boldsymbol{G}_k^i = \frac{\partial h^i(\boldsymbol{x}_k)}{\partial \boldsymbol{x}_k}\big|_{\boldsymbol{x}_k = \hat{\boldsymbol{x}}_{k|k-1}^i}$. Only the first-order term of the expansion of $h^i(\boldsymbol{x}_k)$ is reserved, so one has $\hat{\boldsymbol{z}}_{k|k-1}^i = h^i(\boldsymbol{x}_k) - \boldsymbol{G}_k^i \tilde{\boldsymbol{x}}_{k|k-1}^i$. Since $\boldsymbol{z}_k^i - h^i(\boldsymbol{x}_k) = \boldsymbol{v}_k^i$, Eq. (20) can be rewritten as

$$\tilde{\boldsymbol{x}}_k^i = \boldsymbol{A}_k^i \tilde{\boldsymbol{x}}_{k|k-1}^i - \boldsymbol{M}_k^i \boldsymbol{\kappa}_k^i - \boldsymbol{M}_k^i \boldsymbol{v}_k^i, \tag{22}$$

where $\boldsymbol{A}_k^i = \boldsymbol{I} - \boldsymbol{M}_k^i \boldsymbol{G}_k^i$. Then, the error covariance matrix $\boldsymbol{P}_k^i$ is computed as

$$\begin{aligned} \boldsymbol{P}_k^i =& \mathbb{E}\{\tilde{\boldsymbol{x}}_k^i (\tilde{\boldsymbol{x}}_k^i)^{\mathrm{T}}\} \\ =& \boldsymbol{A}_k^i \boldsymbol{P}_{k|k-1}^i (\boldsymbol{A}_k^i)^{\mathrm{T}} - \boldsymbol{B}_k^i - (\boldsymbol{B}_k^i)^{\mathrm{T}} - \boldsymbol{N}_k^i - (\boldsymbol{N}_k^i)^{\mathrm{T}} \\ & + \boldsymbol{O}_k^i + (\boldsymbol{O}_k^i)^{\mathrm{T}} + \boldsymbol{M}_k^i \mathbb{E}\{\boldsymbol{\kappa}_k^i (\boldsymbol{\kappa}_k^i)^{\mathrm{T}}\} (\boldsymbol{M}_k^i)^{\mathrm{T}} \\ & + \boldsymbol{M}_k^i \mathbb{E}\{\boldsymbol{v}_k^i (\boldsymbol{v}_k^i)^{\mathrm{T}}\} (\boldsymbol{M}_k^i)^{\mathrm{T}}, \end{aligned} \tag{23}$$

where

$$\begin{aligned} \boldsymbol{B}_k^i &= \boldsymbol{A}_k^i \mathbb{E}\{\tilde{\boldsymbol{x}}_{k|k-1}^i (\boldsymbol{\kappa}_k^i)^{\mathrm{T}}\} (\boldsymbol{M}_k^i)^{\mathrm{T}}, \\ \boldsymbol{N}_k^i &= \boldsymbol{A}_k^i \mathbb{E}\{\tilde{\boldsymbol{x}}_{k|k-1}^i (\boldsymbol{v}_k^i)^{\mathrm{T}}\} (\boldsymbol{M}_k^i)^{\mathrm{T}}, \\ \boldsymbol{O}_k^i &= \boldsymbol{M}_k^i \mathbb{E}\{\boldsymbol{\kappa}_k^i (\boldsymbol{v}_k^i)^{\mathrm{T}}\} (\boldsymbol{M}_k^i)^{\mathrm{T}}. \end{aligned} \tag{24}$$

Recall that the measurement noise $\boldsymbol{v}_k^i$ follows a Gaussian distribution with mean 0, it can be demonstrated that $\boldsymbol{N}_k^i = (\boldsymbol{N}_k^i)^{\mathrm{T}} = 0$ since $\tilde{\boldsymbol{x}}_{k|k-1}^i$ and $\boldsymbol{v}_k^i$ are independent from each other. So far, Eq. (23) is equivalent to

$$\begin{aligned} \boldsymbol{P}_k^i =& \boldsymbol{A}_k^i \boldsymbol{P}_{k|k-1}^i (\boldsymbol{A}_k^i)^{\mathrm{T}} - \boldsymbol{B}_k^i - (\boldsymbol{B}_k^i)^{\mathrm{T}} + \boldsymbol{O}_k^i + (\boldsymbol{O}_k^i)^{\mathrm{T}} \\ & + \boldsymbol{M}_k^i \mathbb{E}\{\boldsymbol{\kappa}_k^i (\boldsymbol{\kappa}_k^i)^{\mathrm{T}}\} (\boldsymbol{M}_k^i)^{\mathrm{T}} + \boldsymbol{M}_k^i \boldsymbol{R}_k^i (\boldsymbol{M}_k^i)^{\mathrm{T}}. \end{aligned} \tag{25}$$

From Lemma 2 and

$$\gamma_k^i = \begin{cases} 1, \text{ if } (\boldsymbol{z}_k^i - \boldsymbol{z}_{\tau_{k-1}^i}^i)^{\mathrm{T}} (\boldsymbol{z}_k^i - \boldsymbol{z}_{\tau_{k-1}^i}^i) > \delta^i, \\ 0, \text{ otherwise}, \end{cases} \tag{26}$$

one obtains

$$\begin{aligned} -\boldsymbol{B}_k^i - (\boldsymbol{B}_k^i)^{\mathrm{T}} &\leq \sigma_1 \boldsymbol{A}_k^i \boldsymbol{P}_{k|k-1}^i (\boldsymbol{A}_k^i)^{\mathrm{T}} + \sigma_1^{-1} \boldsymbol{M}_k^i \delta^i (\boldsymbol{M}_k^i)^{\mathrm{T}}, \\ \boldsymbol{O}_k^i + (\boldsymbol{O}_k^i)^{\mathrm{T}} &\leq \sigma_2 \boldsymbol{M}_k^i \boldsymbol{R}_k^i (\boldsymbol{M}_k^i)^{\mathrm{T}} + \sigma_2^{-1} \boldsymbol{M}_k^i \delta^i (\boldsymbol{M}_k^i)^{\mathrm{T}}, \end{aligned} \tag{27}$$

where σ_1 and σ_2 are both positive scalars. Then, the following inequality holds

$$\begin{aligned}\boldsymbol{P}_k^i \leq & \mu_1 \boldsymbol{A}_k^i \boldsymbol{P}_{k|k-1}^i (\boldsymbol{A}_k^i)^{\mathrm{T}} + \mu_2 \boldsymbol{M}_k^i \boldsymbol{R}_k^i (\boldsymbol{M}_k^i)^{\mathrm{T}} \\ & + \mu_3 \boldsymbol{M}_k^i \delta^i (\boldsymbol{M}_k^i)^{\mathrm{T}},\end{aligned} \tag{28}$$

where $\mu_1 = 1+\sigma_1$, $\mu_2 = 1+\sigma_2$, and $\mu_3 = 1+\sigma_1^{-1}+\sigma_2^{-1}$. Therefore, we have the upper bound of the error covariance matrix $\bar{\boldsymbol{P}}_k^i$ as the right-hand side of (28). To further derive the optimal filter gain $\boldsymbol{M}_k^i$, the upper bound $\bar{\boldsymbol{P}}_k^i$ is minimized according to

$$\begin{aligned}\frac{\partial \mathrm{tr}(\bar{\boldsymbol{P}}_k^i)}{\partial \boldsymbol{M}_k^i} = & -2\mu_1(\boldsymbol{I} - \boldsymbol{M}_k^i \boldsymbol{G}_k^i)\boldsymbol{P}_{k|k-1}^i (\boldsymbol{G}_k^i)^{\mathrm{T}} \\ & + 2\mu_2 \boldsymbol{M}_k^i \boldsymbol{R}_k^i + 2\mu_3 \boldsymbol{M}_k^i \delta^i \\ = & \boldsymbol{0},\end{aligned} \tag{29}$$

from which, the optimal filter gain when $\gamma_k^i = 0$ follows

$$\begin{aligned}\boldsymbol{M}_k^i = & \mu_1 \boldsymbol{P}_{k|k-1}^i (\boldsymbol{G}_k^i)^{\mathrm{T}} \\ & \times [\mu_1 \boldsymbol{G}_k^i \boldsymbol{P}_{k|k-1}^i (\boldsymbol{G}_k^i)^{\mathrm{T}} + \mu_2 \boldsymbol{R}_k^i + \mu_3 \delta^i \boldsymbol{I}]^{-1}.\end{aligned} \tag{30}$$

Appendix D
Derivation of ET-CIF

The ET-CIF in non-triggered cases is derived on the basis of ET-CKF as in Appendix B. For convenience, we denote $\boldsymbol{S}_k^i$ as

$$\boldsymbol{S}_k^i = \mu_1 \boldsymbol{G}_k^i \boldsymbol{P}_{k|k-1}^i (\boldsymbol{G}_k^i)^{\mathrm{T}} + \mu_2 \boldsymbol{R}_k^i + \mu_3 \delta^i \boldsymbol{I}. \tag{31}$$

Then, one can rewrite the upper bound of the error covariance matrix $\bar{\boldsymbol{P}}_k^i$ in (28) as

$$\begin{aligned}\bar{\boldsymbol{P}}_k^i = & (1+\sigma_1)(\boldsymbol{I} - \boldsymbol{M}_k^i \boldsymbol{G}_k^i)\boldsymbol{P}_{k|k-1}^i (\boldsymbol{I} - \boldsymbol{M}_k^i \boldsymbol{G}_k^i)^{\mathrm{T}} \\ & + \mu_2 \boldsymbol{M}_k^i \boldsymbol{R}_k^i (\boldsymbol{M}_k^i)^{\mathrm{T}} + \mu_3 \boldsymbol{M}_k^i \delta^i (\boldsymbol{M}_k^i)^{\mathrm{T}} \\ = & \boldsymbol{P}_{k|k-1}^i + \boldsymbol{N}_k^i,\end{aligned} \tag{32}$$

where

$$\begin{aligned}\boldsymbol{N}_k^i = & (\sigma_1 \boldsymbol{I}_{n\times n} - \mu_1 \boldsymbol{M}_k^i \boldsymbol{G}_k^i)\boldsymbol{P}_{k|k-1}^i \\ & - \mu_1 \boldsymbol{P}_{k|k-1}^i (\boldsymbol{M}_k^i \boldsymbol{G}_k^i)^{\mathrm{T}} + \boldsymbol{M}_k^i \boldsymbol{S}_k^i (\boldsymbol{M}_k^i)^{\mathrm{T}}.\end{aligned} \tag{33}$$

For numerical computation reasons, we approximate the error covariance matrix $\boldsymbol{P}_k^i$ to its upper bound $\bar{\boldsymbol{P}}_k^i$. According to Lemma 1, one has

$$(\boldsymbol{A}+\boldsymbol{B})^{-1} = \boldsymbol{A}^{-1} - \boldsymbol{A}^{-1}(\boldsymbol{B}^{-1} + \boldsymbol{A}^{-1})^{-1}\boldsymbol{A}^{-1}. \tag{34}$$

Then, the information matrix is calculated as

$$\begin{aligned}\boldsymbol{Y}_k^i = & (\boldsymbol{P}_{k|k-1}^i + \boldsymbol{N}_k^i)^{-1} \\ = & \boldsymbol{Y}_{k|k-1}^i - \boldsymbol{Y}_{k|k-1}^i [(\boldsymbol{N}_k^i)^{-1} + \boldsymbol{Y}_{k|k-1}^i]^{-1} \boldsymbol{Y}_{k|k-1}^i \\ = & \boldsymbol{Y}_{k|k-1}^i + \boldsymbol{I}_k^i,\end{aligned} \tag{35}$$

from which, the correlation information $\boldsymbol{I}_k^i$ follows

$$\boldsymbol{I}_k^i = -\boldsymbol{Y}_{k|k-1}^i [(\boldsymbol{N}_k^i)^{-1} + \boldsymbol{Y}_{k|k-1}^i]^{-1} \boldsymbol{Y}_{k|k-1}^i. \tag{36}$$

Recall that $\hat{\boldsymbol{x}}_k^i = \hat{\boldsymbol{x}}_{k|k-1}^i + \boldsymbol{M}_k^i(\boldsymbol{z}_{\tau_{k-1}^i}^i - \hat{\boldsymbol{z}}_{k|k-1}^i)$ when $\gamma_k^i = 0$, then the posterior information vector $\hat{\boldsymbol{y}}_k^i$ can be written as

$$\begin{aligned}\hat{\boldsymbol{y}}_k^i = & (\boldsymbol{P}_k^i)^{-1}\hat{\boldsymbol{x}}_k^i \\ = & (\boldsymbol{P}_{k|k-1}^i + \boldsymbol{N}_k^i)^{-1}[\hat{\boldsymbol{x}}_{k|k-1}^i + \boldsymbol{M}_k^i(\boldsymbol{z}_{\tau_{k-1}^i}^i - \hat{\boldsymbol{z}}_{k|k-1}^i)] \\ = & [(\boldsymbol{P}_{k|k-1}^i)^{-1} + \boldsymbol{I}_k^i](\hat{\boldsymbol{x}}_{k|k-1}^i + \boldsymbol{M}_k^i \Delta\bar{\boldsymbol{z}}_k^i) \\ = & \hat{\boldsymbol{y}}_{k|k-1}^i + \mu_1(\boldsymbol{G}_k^i)^{\mathrm{T}}(\boldsymbol{S}_k^i)^{-1}\Delta\bar{\boldsymbol{z}}_k^i + \boldsymbol{I}_k^i(\hat{\boldsymbol{x}}_{k|k-1}^i + \boldsymbol{M}_k^i\Delta\bar{\boldsymbol{z}}_k^i) \\ = & \hat{\boldsymbol{y}}_{k|k-1}^i + \boldsymbol{i}_k^i.\end{aligned} \tag{37}$$

Therefore, the information contribution $\boldsymbol{i}_k^i$ satisfies

$$\boldsymbol{i}_k^i = \mu_1(\boldsymbol{G}_k^i)^{\mathrm{T}}(\boldsymbol{S}_k^i)^{-1}\Delta\bar{\boldsymbol{z}}_k^i + \boldsymbol{I}_k^i(\hat{\boldsymbol{x}}_{k|k-1}^i + \boldsymbol{M}_k^i\Delta\bar{\boldsymbol{z}}_k^i). \tag{38}$$

Appendix E
Stability Analysis

In this section, we analyze the stability of the proposed algorithm by proving the stochastic boundedness of estimation errors in the mean square. Specifically, we prove Theorem 1. To this end, we make the following definitions:

First, the approximation of the nonlinear system in (50) is given for analytical convenience. Define the pseudo system matrix $\boldsymbol{F}_k^i$ in (39) and the pseudo measurement matrix $\boldsymbol{H}_k^i$ in (6) according to [4].

$$\boldsymbol{F}_k^i \triangleq (\boldsymbol{P}_{x_k, x_{k+1|k}}^i)^{\mathrm{T}} \boldsymbol{Y}_k^i. \tag{39}$$

Then, the target tracking model can be linearized as

$$\begin{cases}\boldsymbol{x}_{k+1} = \boldsymbol{\mathcal{A}}_k^i \boldsymbol{F}_k^i \boldsymbol{x}_k + \boldsymbol{w}_k, \\ \boldsymbol{z}_{k+1}^i = \boldsymbol{\mathcal{B}}_{k+1}^i \boldsymbol{H}_{k+1}^i \boldsymbol{x}_{k+1} + \boldsymbol{v}_{k+1}^i, \ \nu_i \in \boldsymbol{\mathcal{V}},\end{cases} \tag{40}$$

where $\boldsymbol{\mathcal{A}}_k^i$ and $\boldsymbol{\mathcal{B}}_k^i$ are compensation instrumental diagonal matrices [5] introduced to neutralize the linearization errors of $\boldsymbol{F}_k^i$ and $\boldsymbol{H}_k^i$, following

$$\boldsymbol{\mathcal{A}}_k^i = \mathrm{diag}\{\alpha_{k,1}^i, \alpha_{k,2}^i, \ldots, \alpha_{k,n}^i\}, \tag{41}$$

$$\boldsymbol{\mathcal{B}}_k^i = \mathrm{diag}\{\beta_{k,1}^i, \beta_{k,2}^i, \ldots, \beta_{k,m}^i\}. \tag{42}$$

Correspondingly, the prior state estimate, predicted measurement, prior error covariance matrix, predicted measurement covariance matrix, and state-measurement cross-covariance matrix satisfy

$$\hat{\boldsymbol{x}}_{k+1|k}^i = \boldsymbol{\mathcal{A}}_k^i \boldsymbol{F}_k^i \hat{\boldsymbol{x}}_{k,\mathrm{fus}}^i, \tag{43}$$

$$\hat{\boldsymbol{z}}_{k+1|k}^i = \boldsymbol{\mathcal{B}}_k^i \boldsymbol{H}_k^i \hat{\boldsymbol{x}}_{k+1|k}^i, \tag{44}$$

$$\boldsymbol{P}_{k+1|k}^i = \boldsymbol{\mathcal{A}}_k^i \boldsymbol{F}_k^i \boldsymbol{P}_{k,\mathrm{fus}}^i (\boldsymbol{\mathcal{A}}_k^i \boldsymbol{F}_k^i)^{\mathrm{T}} + \boldsymbol{Q}_k, \tag{45}$$

$$\boldsymbol{P}_{z_{k+1}, z_{k+1}}^i = \boldsymbol{\mathcal{B}}_{k+1}^i \boldsymbol{H}_{k+1}^i \boldsymbol{P}_{k+1|k}^i (\boldsymbol{\mathcal{B}}_{k+1}^i \boldsymbol{H}_{k+1}^i)^{\mathrm{T}} + \boldsymbol{R}_{k+1}^i, \tag{46}$$

$$\boldsymbol{P}_{x_{k+1}, z_{k+1}}^i = \boldsymbol{P}_{k+1|k}^i (\boldsymbol{\mathcal{B}}_{k+1}^i \boldsymbol{H}_{k+1}^i)^{\mathrm{T}}. \tag{47}$$

Suppose the following reasonable assumptions [6] hold.

Assumption 1. Assume real numbers $\underline{f}, \underline{h}, \underline{a}, \underline{b}, \overline{f}, \overline{h}, \overline{a}, \overline{b} \neq 0$ exist such that the following inequalities are satisfied for all $k \geq 0, \nu_i \in \boldsymbol{\mathcal{V}}$.

$$\begin{cases}\underline{f}^2 \boldsymbol{I}_n \leq \boldsymbol{F}_k^i (\boldsymbol{F}_k^i)^{\mathrm{T}} \leq \overline{f}^2 \boldsymbol{I}_n, \ \underline{h}^2 \boldsymbol{I}_m \leq \boldsymbol{H}_k^i (\boldsymbol{H}_k^i)^{\mathrm{T}} \leq \overline{h}^2 \boldsymbol{I}_m, \\ \underline{a}^2 \boldsymbol{I}_n \leq \boldsymbol{\mathcal{A}}_k^i (\boldsymbol{\mathcal{A}}_k^i)^{\mathrm{T}} \leq \overline{a}^2 \boldsymbol{I}_n, \ \underline{b}^2 \boldsymbol{I}_m \leq \boldsymbol{\mathcal{B}}_k^i (\boldsymbol{\mathcal{B}}_k^i)^{\mathrm{T}} \leq \overline{b}^2 \boldsymbol{I}_m.\end{cases} \tag{48}$$

Assumption 2. Suppose real numbers $p_{\max} \geq p_{\min} > 0, \overline{q} \geq \underline{q} > 0, \overline{r} \geq \underline{r} > 0, \overline{p} \geq \underline{p} > 0$ exist such that the following bounds are fulfilled for all $k \geq 0, \nu_i \in \boldsymbol{\mathcal{V}}$.

$$\begin{cases} p_{\min} \leq p^i \leq p_{\max}, \ \underline{q}\boldsymbol{I}_n \leq \boldsymbol{Q}_k \leq \overline{q}\boldsymbol{I}_n, \\ \underline{r}\boldsymbol{I}_m \leq \boldsymbol{R}_k^i \leq \overline{r}\boldsymbol{I}_m, \ \underline{p}\boldsymbol{I}_n \leq \boldsymbol{P}_{k+1|k}^i \leq \overline{p}\boldsymbol{I}_n. \end{cases} \tag{49}$$

Assumption 3. The undirected network topology is connected.

Based on the aforementioned formulation and assumptions, we are now ready to analyze the stability performance of EDC-CIF.

Theorem 1. *For the nonlinear collaborative tracking system described as*

$$\begin{cases} \boldsymbol{x}_k = f\left(\boldsymbol{x}_{k-1}\right) + \boldsymbol{w}_{k-1}, \\ \boldsymbol{z}_k^i = h^i\left(\boldsymbol{x}_k\right) + \boldsymbol{v}_k^i, \ \nu_i \in \boldsymbol{\mathcal{V}}, \end{cases} \tag{50}$$

the estimation error $\tilde{\boldsymbol{x}}_k^i \triangleq \boldsymbol{x}_k^i - \hat{\boldsymbol{x}}_k^i$ for each $\nu_i \in \boldsymbol{\mathcal{V}}$ at each time instant k in EDC-CIF is exponentially bounded in the mean square under Assumption 1, 2, 3.

Proof. To prove the estimation error's boundedness of EDC-CIF in the mean square, we first introduce the following five necessary lemmas.

Lemma 3 ([7]). *Assume there exist a stochastic process $V(\xi_k)$ and real numbers $\underline{v}, \overline{v}, \mu > 0, 0 < \lambda \leq 1$ such that*

$$\underline{v}\|\xi_k\|^2 \leq V(\xi_k) \leq \overline{v}\|\xi_k\|^2, \tag{51}$$

$$\mathbb{E}\{V(\xi_k)|\xi_{k-1}\} - V(\xi_{k-1}) \leq \mu - \lambda V(\xi_{k-1}), \tag{52}$$

are fulfilled for $\forall k$. Then, the stochastic process's mean square is exponentially bounded, following

$$\mathbb{E}\{\|\xi_k\|^2\} \leq \frac{\overline{v}}{\underline{v}}\mathbb{E}\{\|\xi_0\|^2\}(1-\lambda)^k + \frac{\mu}{\underline{v}}\sum_{i=1}^{k-1}(1-\lambda)^i. \tag{53}$$

Lemma 4 ([8]). *Suppose that matrices $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ and $\boldsymbol{B} \in \mathbb{R}^{n\times n}$. If $\boldsymbol{A} > \boldsymbol{0}$ and $\boldsymbol{B} > \boldsymbol{0}$, then*

$$\boldsymbol{A}^{-1} > (\boldsymbol{A}+\boldsymbol{B})^{-1}. \tag{54}$$

Lemma 5 ([8]). *Give matrices $\boldsymbol{A} \in \mathbb{R}^{m\times m}, \boldsymbol{B} \in \mathbb{R}^{m\times n}$, and $\boldsymbol{C} \in \mathbb{R}^{n\times n}$. If $\boldsymbol{A} > \boldsymbol{0}$ and $\boldsymbol{C} > \boldsymbol{0}$, then*

$$\boldsymbol{A}^{-1} > \boldsymbol{B}(\boldsymbol{B}^{\mathrm{T}}\boldsymbol{A}\boldsymbol{B}+\boldsymbol{C})^{-1}\boldsymbol{B}^{\mathrm{T}}. \tag{55}$$

Lemma 6 ([9]). *Assume there are λ positive definite matrices $\boldsymbol{M}_1, \boldsymbol{M}_2, ..., \boldsymbol{M}_\lambda$ and λ vectors $\boldsymbol{v}_1, \boldsymbol{v}_2, ..., \boldsymbol{v}_\lambda$ with an integer $\lambda \geq 2$. The following inequality holds*

$$(\sum_{i=1}^{\lambda}\boldsymbol{M}_i\boldsymbol{v}_i)^{\mathrm{T}}(\sum_{i=1}^{\lambda}\boldsymbol{M}_i)^{-1}(\sum_{i=1}^{\lambda}\boldsymbol{M}_i\boldsymbol{v}_i) \leq \sum_{i=1}^{\lambda}\boldsymbol{v}_i^{\mathrm{T}}\boldsymbol{M}_i\boldsymbol{v}_i. \tag{56}$$

Lemma 7 ([10]). *Given a set of vectors $\{\boldsymbol{v}_i\}_{i=1}^{\lambda}$ with a positive integer λ, a set of non-negative scalars $\{p_i\}_{i=1}^{\lambda}$ summing up to one, and a positive definite matrix $\boldsymbol{M}$, the following inequality holds*

$$(\sum_{i=1}^{\lambda}p_i\boldsymbol{v}_i)^{\mathrm{T}}\boldsymbol{M}(\sum_{i=1}^{\lambda}p_i\boldsymbol{v}_i) \leq \sum_{i=1}^{\lambda}p_i\boldsymbol{v}_i^{\mathrm{T}}\boldsymbol{M}\boldsymbol{v}_i. \tag{57}$$

Next, the proof of Theorem 1 is presented. Since EDC-CIF is the information equivalence of the corresponding CKF-based algorithm, we conduct the proof based on the estimation pair, including the state estimate and error covariance matrix instead of the information pair, for simplicity. Recall the local prediction error $\tilde{\boldsymbol{x}}_{k|k-1}^i$ and estimation error $\tilde{\boldsymbol{x}}_k^i$ given in (17). According to (18), (77), and the linearized target tracking model

$$\begin{cases} \boldsymbol{x}_{k+1} = \boldsymbol{\mathcal{A}}_k^i\boldsymbol{F}_k^i\boldsymbol{x}_k + \boldsymbol{w}_k, \\ \boldsymbol{z}_{k+1}^i = \boldsymbol{\mathcal{B}}_{k+1}^i\boldsymbol{H}_{k+1}^i\boldsymbol{x}_{k+1} + \boldsymbol{v}_{k+1}^i, \ \nu_i \in \boldsymbol{\mathcal{V}}, \end{cases} \tag{58}$$

it follows that

$$\begin{aligned} \tilde{\boldsymbol{x}}_{k+1|k}^i =& \boldsymbol{x}_{k+1} - \hat{\boldsymbol{x}}_{k+1|k}^i \\ =& \boldsymbol{\mathcal{A}}_k^i\boldsymbol{F}_k^i(\boldsymbol{x}_k - \boldsymbol{x}_{k,\text{fus}}^i) + \boldsymbol{w}_k \\ =& \boldsymbol{\mathcal{A}}_k^i\boldsymbol{F}_k^i\boldsymbol{P}_{k,\text{fus}}^i\sum_{\nu_j\in\boldsymbol{\mathcal{V}}}c_k^{i,j}(\boldsymbol{P}_k^j)^{-1}(\boldsymbol{x}_k - \hat{\boldsymbol{x}}_k^j) + \boldsymbol{w}_k \\ =& \boldsymbol{\mathcal{A}}_k^i\boldsymbol{F}_k^i\boldsymbol{P}_{k,\text{fus}}^i\sum_{\nu_j\in\boldsymbol{\mathcal{V}}}c_k^{i,j}(\boldsymbol{P}_k^j)^{-1}[\boldsymbol{x}_k - \hat{\boldsymbol{x}}_{k|k-1}^j \\ & -\gamma_k^j\boldsymbol{K}_k^j(\boldsymbol{z}_k^j - \hat{\boldsymbol{z}}_{k|k-1}^j) - (1-\gamma_k^j)\boldsymbol{M}_k^j(\boldsymbol{z}_{\tau_{k-1}^j}^j - \hat{\boldsymbol{z}}_{k|k-1}^j)] \\ & + \boldsymbol{w}_k \\ =& \sum_{\nu_j\in\boldsymbol{\mathcal{V}}}\boldsymbol{\Omega}_k^{i,j}\tilde{\boldsymbol{x}}_{k|k-1}^j + \sum_{\nu_j\in\boldsymbol{\mathcal{V}}}\boldsymbol{\Lambda}_k^{i,j}(\boldsymbol{z}_k^j - \boldsymbol{z}_{\tau_{k-1}^j}^j) \\ & + \sum_{\nu_j\in\boldsymbol{\mathcal{V}}}\boldsymbol{\Theta}_k^{i,j}\boldsymbol{v}_k^j + \boldsymbol{w}_k, \end{aligned} \tag{59}$$

where

$$\begin{cases} \boldsymbol{\Omega}_k^{i,j} =& c_k^{i,j}\boldsymbol{\mathcal{A}}_k^i\boldsymbol{F}_k^i\boldsymbol{P}_{k,\text{fus}}^i(\boldsymbol{P}_k^j)^{-1} \\ & \times[\boldsymbol{I} - \gamma_k^j\boldsymbol{K}_k^j\boldsymbol{\mathcal{B}}_k^j\boldsymbol{H}_k^j - (1-\gamma_k^j)\boldsymbol{M}_k^j\boldsymbol{\mathcal{B}}_k^j\boldsymbol{H}_k^j], \\ \boldsymbol{\Lambda}_k^{i,j} =& c_k^{i,j}\boldsymbol{\mathcal{A}}_k^i\boldsymbol{F}_k^i\boldsymbol{P}_{k,\text{fus}}^i(\boldsymbol{P}_k^j)^{-1}(1-\gamma_k^j)\boldsymbol{M}_k^j, \\ \boldsymbol{\Theta}_k^{i,j} =& c_k^{i,j}\boldsymbol{\mathcal{A}}_k^i\boldsymbol{F}_k^i\boldsymbol{P}_{k,\text{fus}}^i(\boldsymbol{P}_k^j)^{-1}[-\gamma_k^j\boldsymbol{K}_k^j - (1-\gamma_k^j)\boldsymbol{M}_k^j]. \end{cases} \tag{60}$$

Denote $\tilde{\boldsymbol{x}}_{k+1|k} = \text{col}(\tilde{\boldsymbol{x}}_{k+1|k}^i, \nu_i \in \boldsymbol{\mathcal{V}})$ and $\tilde{\boldsymbol{x}}_k = \text{col}(\tilde{\boldsymbol{x}}_k^i, \nu_i \in \boldsymbol{\mathcal{V}})$. let vector $\boldsymbol{p} = [p^1, \ldots, p^N]^{\mathrm{T}}$ represent the Perron-Frobenius left eigenvector of the diffusion weight matrix $\boldsymbol{C}_k = [c_k^{i,j}]_{N\times N}$. From Assumption 3, we have $\sum_{\nu_j\in\boldsymbol{\mathcal{V}}}p^ic_k^{i,j} = p^j$. Consider the stochastic process as

$$V(\tilde{\boldsymbol{x}}_{k+1|k}) = \sum_{\nu_i\in\boldsymbol{\mathcal{V}}}p^i(\tilde{\boldsymbol{x}}_{k+1|k}^i)^{\mathrm{T}}(\boldsymbol{P}_{k+1|k}^i)^{-1}\tilde{\boldsymbol{x}}_{k+1|k}^i. \tag{61}$$

According to $\underline{p}\boldsymbol{I}_n \leq \boldsymbol{P}_{k+1|k}^i \leq \overline{p}\boldsymbol{I}_n$ in Assumption 2, it can be obtained that

$$\frac{p_{\min}}{\overline{p}}\|\tilde{\boldsymbol{x}}_{k+1|k}\|^2 \leq V(\tilde{\boldsymbol{x}}_{k+1|k}) \leq \frac{p_{\max}}{\underline{p}}\|\tilde{\boldsymbol{x}}_{k+1|k}\|^2, \tag{62}$$

which meets the condition (51) in Lemma 3. For convenience, we further define

$$\boldsymbol{\mathcal{R}}_k^{i,j} = \sum_{\nu_j\in\boldsymbol{\mathcal{V}}}\boldsymbol{\Omega}_k^{i,j}\tilde{\boldsymbol{x}}_{k|k-1}^j, \tag{63}$$

$$\boldsymbol{\mathcal{S}}_k^{i,j} = \sum_{\nu_j\in\boldsymbol{\mathcal{V}}}\boldsymbol{\Lambda}_k^{i,j}(\boldsymbol{z}_k^j - \boldsymbol{z}_{\tau_{k-1}^j}^j), \tag{64}$$

$$\boldsymbol{\Gamma}_k^{i,j} = \sum_{\nu_j \in \boldsymbol{\mathcal{V}}} \boldsymbol{\Theta}_k^{i,j} \boldsymbol{v}_k^j + \boldsymbol{w}_k = \boldsymbol{\mathcal{U}}_k^{i,j} + \boldsymbol{w}_k. \tag{65}$$

Then, the conditional expectation of (61) can be written as

$$\begin{aligned} &\mathbb{E}\{V(\tilde{\boldsymbol{x}}_{k+1|k})|\tilde{\boldsymbol{x}}_{k|k-1}\} \\ =&\mathbb{E}\{\sum_{\nu_i \in \boldsymbol{\mathcal{V}}} p^i (\tilde{\boldsymbol{x}}_{k+1|k}^i)^{\mathrm{T}} (\boldsymbol{P}_{k+1|k}^i)^{-1} \tilde{\boldsymbol{x}}_{k+1|k}^i |\tilde{\boldsymbol{x}}_{k|k-1}\}\} \\ =&\psi_{k+1}^x + \psi_{k+1}^z + \psi_{k+1}^{v,w} + \psi_{k+1}^{x,z} + \psi_{k+1}^{z,v,w}, \end{aligned} \tag{66}$$

where

$$\psi_{k+1}^x = \mathbb{E}\{\sum_{\nu_i \in \boldsymbol{\mathcal{V}}} p^i (\boldsymbol{\mathcal{R}}_k^{i,j})^{\mathrm{T}} (\boldsymbol{P}_{k+1|k}^i)^{-1} \boldsymbol{\mathcal{R}}_k^{i,j} |\tilde{\boldsymbol{x}}_{k|k-1}\}, \tag{67}$$

$$\psi_{k+1}^z = \mathbb{E}\{\sum_{\nu_i \in \boldsymbol{\mathcal{V}}} p^i (\boldsymbol{\mathcal{S}}_k^{i,j})^{\mathrm{T}} (\boldsymbol{P}_{k+1|k}^i)^{-1} \boldsymbol{\mathcal{S}}_k^{i,j} |\tilde{\boldsymbol{x}}_{k|k-1}\}, \tag{68}$$

$$\psi_{k+1}^{v,w} = \mathbb{E}\{\sum_{\nu_i \in \boldsymbol{\mathcal{V}}} p^i (\boldsymbol{\Gamma}_k^{i,j})^{\mathrm{T}} (\boldsymbol{P}_{k+1|k}^i)^{-1} \boldsymbol{\Gamma}_k^{i,j} |\tilde{\boldsymbol{x}}_{k|k-1}\}, \tag{69}$$

$$\begin{aligned} \psi_{k+1}^{x,z} =&\mathbb{E}\{\sum_{\nu_i \in \boldsymbol{\mathcal{V}}} p^i [(\boldsymbol{\mathcal{R}}_k^{i,j})^{\mathrm{T}} (\boldsymbol{P}_{k+1|k}^i)^{-1} \boldsymbol{\mathcal{S}}_k^{i,j} + (\boldsymbol{\mathcal{S}}_k^{i,j})^{\mathrm{T}} \\ &\times (\boldsymbol{P}_{k+1|k}^i)^{-1} \boldsymbol{\mathcal{R}}_k^{i,j}] |\tilde{\boldsymbol{x}}_{k|k-1}\}, \end{aligned} \tag{70}$$

$$\begin{aligned} \psi_{k+1}^{z,v,w} =& \mathbb{E}\{\sum_{\nu_i \in \boldsymbol{\mathcal{V}}} p^i [(\boldsymbol{\mathcal{S}}_k^{i,j})^{\mathrm{T}} (\boldsymbol{P}_{k+1|k}^i)^{-1} \boldsymbol{\Gamma}_k^{i,j} + (\boldsymbol{\Gamma}_k^{i,j})^{\mathrm{T}} \\ &\times (\boldsymbol{P}_{k+1|k}^i)^{-1} \boldsymbol{\mathcal{S}}_k^{i,j}] |\tilde{\boldsymbol{x}}_{k|k-1}\}, \end{aligned} \tag{71}$$

The five expectation terms in (66) are discussed next to prove the boundedness of the stochastic process. Since the homogeneous recursion greatly affects the algorithm's stability, we first focus on the term ψ_{k+1}^x. According to Lemma 4 and $\boldsymbol{P}_{k+1|k}^i = \boldsymbol{\mathcal{A}}_k^i \boldsymbol{F}_k^i \boldsymbol{P}_{k,\text{fus}}^i (\boldsymbol{\mathcal{A}}_k^i \boldsymbol{F}_k^i)^{\mathrm{T}} + \boldsymbol{Q}_k$, one has

$$(\boldsymbol{P}_{k+1|k}^i)^{-1} \leq (\boldsymbol{\mathcal{A}}_k^i \boldsymbol{\mathcal{F}}_k^i)^{-\mathrm{T}} (\boldsymbol{P}_{k,\text{fus}}^i)^{-1} (\boldsymbol{\mathcal{A}}_k^i \boldsymbol{\mathcal{F}}_k^i)^{-1}. \tag{72}$$

From Assumption 1, it can be guaranteed that $\boldsymbol{\mathcal{A}}_k^i \boldsymbol{\mathcal{F}}_k^i$ is non-singular. Next, substituting (72) into (67) yields

$$\begin{aligned} \psi_{k+1}^x \leq&\mathbb{E}\{\sum_{\nu_i \in \boldsymbol{\mathcal{V}}} p^i [\boldsymbol{P}_{k,\text{fus}}^i \sum_{\nu_j \in \boldsymbol{\mathcal{V}}} c_k^{i,j} (\boldsymbol{P}_k^j)^{-1} \boldsymbol{\Upsilon}_k^j \tilde{\boldsymbol{x}}_{k|k-1}^j]^{\mathrm{T}} \\ &\times \sum_{\nu_j \in \boldsymbol{\mathcal{V}}} c_k^{i,j} (\boldsymbol{P}_k^j)^{-1} \boldsymbol{\Upsilon}_k^j \tilde{\boldsymbol{x}}_{k|k-1}^j |\tilde{\boldsymbol{x}}_{k|k-1}\}, \end{aligned} \tag{73}$$

where $\boldsymbol{\Upsilon}_k^j = \boldsymbol{I} - \gamma_k^j \boldsymbol{K}_k^j \boldsymbol{\mathcal{B}}_k^j \boldsymbol{H}_k^j - (1-\gamma_k^j) \boldsymbol{M}_k^j \boldsymbol{\mathcal{B}}_k^j \boldsymbol{H}_k^j$. Recalling that $(\boldsymbol{P}_{k,\text{fus}}^i)^{-1} = \sum_{\nu_j \in \boldsymbol{\mathcal{V}}} c_k^{i,j} (\boldsymbol{P}_k^j)^{-1}$, one can rewrite (73) as

$$\begin{aligned} \psi_{k+1}^x \leq&\mathbb{E}\{\sum_{\nu_i \in \boldsymbol{\mathcal{V}}} p^i [\sum_{\nu_j \in \boldsymbol{\mathcal{V}}} c_k^{i,j} (\boldsymbol{P}_k^j)^{-1} \boldsymbol{\Upsilon}_k^j \tilde{\boldsymbol{x}}_{k|k-1}^j]^{\mathrm{T}} \\ &\times [\sum_{\nu_j \in \boldsymbol{\mathcal{V}}} c_k^{i,j} (\boldsymbol{P}_k^j)^{-1}]^{-1} \\ &\times \sum_{\nu_j \in \boldsymbol{\mathcal{V}}} c_k^{i,j} (\boldsymbol{P}_k^j)^{-1} \boldsymbol{\Upsilon}_k^j \tilde{\boldsymbol{x}}_{k|k-1}^j |\tilde{\boldsymbol{x}}_{k|k-1}\}. \end{aligned} \tag{74}$$

Note that $\sum_{\nu_i \in \boldsymbol{\mathcal{V}}} p^i c_k^{i,j} = p^j$. Then, applying Lemma 6 to (74) gives

$$\begin{aligned} \psi_{k+1}^x \leq&\mathbb{E}\{\sum_{\nu_i \in \boldsymbol{\mathcal{V}}} p^i \sum_{\nu_j \in \boldsymbol{\mathcal{V}}} (\boldsymbol{\Upsilon}_k^j \tilde{\boldsymbol{x}}_{k|k-1}^j)^{\mathrm{T}} c_k^{i,j} (\boldsymbol{P}_k^j)^{-1} \\ &\times \boldsymbol{\Upsilon}_k^j \tilde{\boldsymbol{x}}_{k|k-1}^j |\tilde{\boldsymbol{x}}_{k|k-1}\} \\ =&\mathbb{E}\{\sum_{\nu_j \in \boldsymbol{\mathcal{V}}} p^j \sum_{\nu_j \in \boldsymbol{\mathcal{V}}} (\boldsymbol{\Upsilon}_k^j \tilde{\boldsymbol{x}}_{k|k-1}^j)^{\mathrm{T}} (\boldsymbol{P}_k^j)^{-1} \\ &\times \boldsymbol{\Upsilon}_k^j \tilde{\boldsymbol{x}}_{k|k-1}^j |\tilde{\boldsymbol{x}}_{k|k-1}\}. \end{aligned} \tag{75}$$

Recall the linearized prior state estimate, predicted measurement, prior error covariance matrix, predicted measurement covariance matrix, and state-measurement cross-covariance matrix as

$$\hat{\boldsymbol{x}}_{k+1|k}^i = \boldsymbol{\mathcal{A}}_k^i \boldsymbol{F}_k^i \hat{\boldsymbol{x}}_{k,\text{fus}}^i, \tag{76}$$

$$\hat{\boldsymbol{z}}_{k+1|k}^i = \boldsymbol{\mathcal{B}}_k^i \boldsymbol{H}_k^i \hat{\boldsymbol{x}}_{k+1|k}^i, \tag{77}$$

$$\boldsymbol{P}_{k+1|k}^i = \boldsymbol{\mathcal{A}}_k^i \boldsymbol{F}_k^i \boldsymbol{P}_{k,\text{fus}}^i (\boldsymbol{\mathcal{A}}_k^i \boldsymbol{F}_k^i)^{\mathrm{T}} + \boldsymbol{Q}_k, \tag{78}$$

$$\boldsymbol{P}_{z_{k+1},z_{k+1}}^i = \boldsymbol{\mathcal{B}}_{k+1}^i \boldsymbol{H}_{k+1}^i \boldsymbol{P}_{k+1|k}^i (\boldsymbol{\mathcal{B}}_{k+1}^i \boldsymbol{H}_{k+1}^i)^{\mathrm{T}} + \boldsymbol{R}_{k+1}^i, \tag{79}$$

$$\boldsymbol{P}_{x_{k+1},z_{k+1}}^i = \boldsymbol{P}_{k+1|k}^i (\boldsymbol{\mathcal{B}}_{k+1}^i \boldsymbol{H}_{k+1}^i)^{\mathrm{T}}. \tag{80}$$

According to $\bar{\boldsymbol{P}}_k^i = \gamma_k^i [\boldsymbol{P}_{k|k-1}^i - \boldsymbol{K}_k^i \boldsymbol{P}_{z_k,z_k}^i (\boldsymbol{K}_k^i)^{\mathrm{T}}] + (1-\gamma_k^i) \bar{\boldsymbol{P}}_k^i$, (79), (80), and (30), we rewrite (75) as

$$\begin{aligned} \psi_{k+1}^x \leq& \mathbb{E}\{\sum_{\nu_j \in \boldsymbol{\mathcal{V}}} p^j \sum_{\nu_j \in \boldsymbol{\mathcal{V}}} (\tilde{\boldsymbol{x}}_{k|k-1}^j)^{\mathrm{T}} (\boldsymbol{P}_{k|k-1}^j)^{-1} \\ &\times \boldsymbol{\Upsilon}_k^j \tilde{\boldsymbol{x}}_{k|k-1}^j |\tilde{\boldsymbol{x}}_{k|k-1}\} \\ =& \mathbb{E}\{\sum_{\nu_j \in \boldsymbol{\mathcal{V}}} p^j (\tilde{\boldsymbol{x}}_{k|k-1}^j)^{\mathrm{T}} (\boldsymbol{P}_{k|k-1}^j)^{-1} \tilde{\boldsymbol{x}}_{k|k-1}^j |\tilde{\boldsymbol{x}}_{k|k-1}\} \\ &-\mathbb{E}\{\sum_{\nu_j \in \boldsymbol{\mathcal{V}}} p^j (\tilde{\boldsymbol{x}}_{k|k-1}^j)^{\mathrm{T}} (\boldsymbol{P}_{k|k-1}^j)^{-1} \\ &\times [\gamma_k^j \boldsymbol{K}_k^j \boldsymbol{\mathcal{B}}_k^j \boldsymbol{H}_k^j + (1-\gamma_k^j) \boldsymbol{M}_k^j \boldsymbol{\mathcal{B}}_k^j \boldsymbol{H}_k^j] \\ &\times \tilde{\boldsymbol{x}}_{k|k-1}^j |\tilde{\boldsymbol{x}}_{k|k-1}\}. \end{aligned} \tag{81}$$

For the second term of (81), we first consider the case where $\gamma_k^j = 1$. Based on Lemma 5, substituting (79) and (80) into $\boldsymbol{K}_k^i = \boldsymbol{P}_{x_k,z_k}^i (\boldsymbol{P}_{z_k,z_k}^i)^{-1}$ and updating (81) with the updated expression of $\boldsymbol{K}_k^j$, one can find a real number λ satisfying $0 < \lambda < \lambda_1 \leq 1$, where

$$\begin{aligned} \lambda_1 =& \min\{\pi_{\min}\{(\boldsymbol{\mathcal{B}}_s^j \boldsymbol{H}_s^j)^{\mathrm{T}} [\boldsymbol{\mathcal{B}}_s^j \boldsymbol{H}_s^j \boldsymbol{P}_{s|s-1}^j (\boldsymbol{\mathcal{B}}_s^j \boldsymbol{H}_s^j)^{\mathrm{T}} + \boldsymbol{R}_s^j]^{-1} \\ &\times \boldsymbol{\mathcal{B}}_s^j \boldsymbol{H}_s^j\}/\pi_{\max}[(\boldsymbol{P}_{s|s-1}^j)^{-1}], \nu_j \in \boldsymbol{\mathcal{V}}, s = 0, 1, \ldots, k\}, \end{aligned} \tag{82}$$

where $\pi_{\min}$ and $\pi_{\max}$ represent the matrix's minimum and maximum eigenvalue, respectively, such that

$$\begin{aligned} \lambda (\boldsymbol{P}_{k|k-1}^j)^{-1} \leq& (\boldsymbol{\mathcal{B}}_k^j \boldsymbol{H}_k^j)^{\mathrm{T}} [\boldsymbol{\mathcal{B}}_k^j \boldsymbol{H}_k^j \boldsymbol{P}_{k|k-1}^j (\boldsymbol{\mathcal{B}}_k^j \boldsymbol{H}_k^j)^{\mathrm{T}} + \boldsymbol{R}_k^j]^{-1} \\ &\times \boldsymbol{\mathcal{B}}_k^j \boldsymbol{H}_k^j, \end{aligned} \tag{83}$$

which is in accordance with the results in [11] and [12]. Similarly, for the case where $\gamma_k^j = 0$, applying (30) into (81),

there also exists a real number λ following $0<\lambda<\lambda_2\leq 1$, where

$$\begin{aligned}\lambda_2 =& \min\{\pi_{\min}\{(\boldsymbol{\mathcal{B}}_s^j\boldsymbol{H}_s^j)^{\mathrm{T}}[\mu_1\boldsymbol{\mathcal{B}}_s^j\boldsymbol{H}_s^j\boldsymbol{P}_{s|s-1}^j(\boldsymbol{\mathcal{B}}_s^j\boldsymbol{H}_s^j)^{\mathrm{T}}+\mu_2\boldsymbol{R}_s^j\\ &+\mu_3\delta^j\boldsymbol{I}]^{-1}\boldsymbol{\mathcal{B}}_s^j\boldsymbol{H}_s^j\}/\pi_{\max}\{[(1+\mu_1)\boldsymbol{P}_{s|s-1}^j]^{-1}\},\\ &\nu_j\in\boldsymbol{\mathcal{V}}, s=0,1,\ldots,k\},\end{aligned} \tag{84}$$

such that

$$\begin{aligned}\lambda(\mu_1\boldsymbol{P}_{k|k-1}^j)^{-1}\leq&(\boldsymbol{\mathcal{B}}_k^j\boldsymbol{H}_k^j)^{\mathrm{T}}[\mu_1\boldsymbol{\mathcal{B}}_k^j\boldsymbol{H}_k^j\boldsymbol{P}_{k|k-1}^j(\boldsymbol{\mathcal{B}}_k^j\boldsymbol{H}_k^j)^{\mathrm{T}}\\ &+\mu_2\boldsymbol{R}_k^j+\mu_3\delta^j\boldsymbol{I}]^{-1}\boldsymbol{\mathcal{B}}_k^j\boldsymbol{H}_k^j.\end{aligned} \tag{85}$$

Let $\lambda^*=\min\{\lambda_1,\lambda_2\}$ and $0<\lambda<\lambda^*\leq 1$. Then, Eq. (81) can be further written as

$$\psi_{k+1}^x\leq(1-\lambda)\mathbb{E}\{V(\tilde{\boldsymbol{x}}_{k|k-1})\}. \tag{86}$$

As for the second expectation term ψ_{k+1}^z in (66), considering Lemma 7, one gets

$$\begin{aligned}\psi_{k+1}^z\leq&\frac{\overline{a}^2\overline{f}^2\overline{m}^2}{\underline{p}}\mathbb{E}\{\sum_{\nu_i\in\boldsymbol{\mathcal{V}}}p^i\sum_{\nu_j\in\boldsymbol{\mathcal{V}}}c_k^{i,j}(1-\gamma_k^j)(\boldsymbol{z}_k^j-\boldsymbol{z}_{\tau_{k-1}^j}^j)^{\mathrm{T}}\\ &\times(\boldsymbol{z}_k^j-\boldsymbol{z}_{\tau_{k-1}^j}^j)|\tilde{\boldsymbol{x}}_{k|k-1})\}\\ \leq&\frac{\overline{a}^2\overline{f}^2\overline{m}^2}{\underline{p}}\sum_{\nu_j\in\boldsymbol{\mathcal{V}}}p^j\delta^j,\end{aligned} \tag{87}$$

where $\overline{m}=\frac{\sigma_1\overline{p}\overline{h}}{\sigma_1\underline{p}\underline{h}^2+\sigma_2\underline{r}+\sigma_3\underline{\delta}}$, and $\underline{\delta}$ denotes the minimum trigger threshold.

Regarding the third expectation term $\psi_{k+1}^{v,w}$, it follows that

$$\begin{aligned}&\psi_{k+1}^{v,w}\\ \leq&\frac{1}{\underline{p}}\mathbb{E}\{\sum_{\nu_i\in\boldsymbol{\mathcal{V}}}p^i[\mathrm{tr}\{(\boldsymbol{\mathcal{U}}_k^{i,j})^{\mathrm{T}}\boldsymbol{\mathcal{U}}_k^{i,j}\}+\mathrm{tr}\{(\boldsymbol{w}_k^{i,j})^{\mathrm{T}}\boldsymbol{w}_k^{i,j}\}]|\tilde{\boldsymbol{x}}_{k|k-1})\}\\ \leq&\frac{1}{\underline{p}}\{\mathbb{E}\{\sum_{\nu_i\in\boldsymbol{\mathcal{V}}}p^i[\overline{a}^2\overline{f}^2\overline{r}(\overline{k}^2+\overline{m}^2)n\sum_{\nu_j\in\boldsymbol{\mathcal{V}}}c_k^{i,j}+\overline{q}n]\}\\ =&\frac{[\overline{a}^2\overline{f}^2\overline{r}(\overline{k}^2+\overline{m}^2)+\overline{q}]n}{\underline{p}}\sum_{\nu_j\in\boldsymbol{\mathcal{V}}}p^j,\end{aligned} \tag{88}$$

where $\overline{k}=\frac{\overline{p}\overline{b}\overline{h}}{\underline{p}\underline{b}^2\underline{h}^2+\underline{r}}$.

In terms of the fourth term $\psi_{k+1}^{x,z}$, applying Lemmas 2 and 7 results in

$$\begin{aligned}\psi_{k+1}^{x,z}\leq&\frac{1}{\underline{p}}\mathbb{E}\{\sum_{\nu_i\in\boldsymbol{\mathcal{V}}}p^i\mathrm{tr}\{\sigma_3\boldsymbol{\mathcal{R}}_k^{i,j}(\boldsymbol{\mathcal{R}}_k^{i,j})^{\mathrm{T}}+\sigma_3^{-1}\boldsymbol{\mathcal{S}}_k^{i,j}(\boldsymbol{\mathcal{S}}_k^{i,j})^{\mathrm{T}}\}|\tilde{\boldsymbol{x}}_{k|k-1})\}\\ \leq&\frac{\sigma_3\overline{p}\overline{a}^2\overline{f}^2(1+\overline{kbh}+\overline{m}\overline{h})^2n}{\underline{p}}\sum_{\nu_j\in\boldsymbol{\mathcal{V}}}p^j\\ &+\frac{\sigma_3^{-1}\overline{a}^2\overline{f}^2\overline{m}^2}{\underline{p}}\sum_{\nu_j\in\boldsymbol{\mathcal{V}}}p^j\delta^j.\end{aligned} \tag{89}$$

Similarly, concerning the last term $\psi_{k+1}^{z,v,w}$, we have

$$\begin{aligned}\psi_{k+1}^{z,v,w}\leq&\frac{\sigma_4[\overline{a}^2\overline{f}^2\overline{r}(\overline{k}^2+\overline{m}^2)+\overline{q}]n}{\underline{p}}\sum_{\nu_j\in\boldsymbol{\mathcal{V}}}p^j\\ &+\frac{\sigma_4^{-1}\overline{a}^2\overline{f}^2\overline{m}^2}{\underline{p}}\sum_{\nu_j\in\boldsymbol{\mathcal{V}}}p^j\delta^j,\end{aligned} \tag{90}$$

where σ_3 and σ_4 are both positive scalars. Define the value of μ as

$$\begin{aligned}\mu\triangleq&\Big\{\frac{(1+\sigma_4)[\overline{a}^2\overline{f}^2\overline{r}(\overline{k}^2+\overline{m}^2)+\overline{q}]n}{\underline{p}}\\ &+\frac{\sigma_3\overline{p}\overline{a}^2\overline{f}^2(1+\overline{kbh}+\overline{m}\overline{h})^2n}{\underline{p}}\Big\}\sum_{\nu_j\in\boldsymbol{\mathcal{V}}}p^j\\ &+\frac{(1+\sigma_3^{-1}+\sigma_4^{-1})\overline{a}^2\overline{f}^2\overline{m}^2}{\underline{p}}\sum_{\nu_j\in\boldsymbol{\mathcal{V}}}p^j\delta^j.\end{aligned} \tag{91}$$

So far, condition (52) in Lemma 3 has been fulfilled. Therefore, it can be deduced that the prediction error $\tilde{\boldsymbol{x}}_{k+1|k}$ is exponentially bounded in the mean square. Then, for each node $\nu_i\in\boldsymbol{\mathcal{V}}$ in the sensor network, the mean square of each component of $\tilde{\boldsymbol{x}}_{k+1|k}$, i.e., the local prediction error $\tilde{\boldsymbol{x}}_{k+1|k}^i$, is also exponentially bounded. For further analysis of the stochastic boundedness of each node's estimation error $\tilde{\boldsymbol{x}}_k^i=\tilde{\boldsymbol{x}}_{k,\mathrm{fus}}^i$, taking expectation from both sides of $\tilde{\boldsymbol{x}}_{k+1|k}^i=\boldsymbol{\mathcal{A}}_k^i\boldsymbol{F}_k^i\tilde{\boldsymbol{x}}_k^i+\boldsymbol{w}_k$ yields

$$\mathbb{E}\{\|\tilde{\boldsymbol{x}}_k^i\|^2\}\leq\underline{a}^{-2}\underline{f}^{-2}\Big(\mathbb{E}\{\|\tilde{\boldsymbol{x}}_{k+1|k}^i\|^2\}-\mathbb{E}\{\|\boldsymbol{w}_k\|^2\}\Big). \tag{92}$$

As the mean square of the prediction error $\tilde{\boldsymbol{x}}_{k+1|k}^i$ and process noise $\boldsymbol{w}_k$ are both exponentially bounded, it can be concluded that the estimation error $\tilde{\boldsymbol{x}}_k^i$ of EDC-CIF is also exponentially bounded in mean square for each node at each time step.

□

Appendix F
Ablation Study on Designed Modules and Alternative Communication-efficient Method

To thoroughly validate the contributions of our proposed EDC-CIF framework, we conduct ablation studies on its two core modules and compare it against an alternative communication-efficient method. Specifically, We evaluate the individual contributions of the error-minimized ET filter and the correlation-aware diffusion module by removing them separately, and compare performance with a dynamic ET mechanism. Results in Table I highlight the contribution of each module and demonstrate the overall advantages of our approach.

TABLE I
Module ablation studies and alternative ET mechanism comparison for EDC-CIF.

Method	$\mathrm{RMSE}_{\mathrm{pos}}$ (m)	$\mathrm{RMSE}_{\mathrm{vel}}$ (m/s)	t_{com} (ms)	TR (%)
EDC-CIF (w/o optimal ET filter)	0.0951	0.3995	2.58	52.33
EDC-CIF (w/o diffusion with CI)	0.0841	0.3759	1.57	52.33
DEC-CKF (dynamic ET)	0.1326	0.5936	2.47	**43.25**
EDC-CIF (ours)	**0.0639**	**0.2919**	**1.32**	52.33

TABLE II
RMSE OF THE STATE VARIABLE ESTIMATION WITH DIFFERENT OBSERVER NUMBERS FOR ALL BASELINES.

Estimation Error		**Non-ET schemes**			**ET schemes** (communication-efficient filters)				
		C-CKF	D-CIF	DC-CIF	EC-EKF	EC-UKF	EC-CKF	ED-CIF	EDC-CIF
3 observers	x (m)	0.0558	0.0536	0.0533	0.3973	0.8943	0.0781	0.0604	**0.0531**
	y (m)	0.2651	0.2723	0.2723	2.0307	5.5478	0.3784	0.2997	**0.2625**
	z (m)	0.0451	0.0449	0.0448	0.3563	0.6311	0.0676	0.0518	**0.0445**
	$\dot{x}$ (m/s)	0.2173	0.2236	0.2215	2.0548	5.2400	0.3200	0.2468	**0.2155**
	$\dot{y}$ (m/s)	0.0513	0.0501	0.0498	0.2412	0.4045	0.0743	0.0568	**0.0496**
	$\dot{z}$ (m/s)	0.1403	0.1425	0.1377	0.3084	0.9672	0.1737	0.1520	**0.1357**
	ω (rad/s)	0.1177	0.1253	0.1231	0.5943	0.5246	0.1352	0.1270	**0.1179**
4 observers	x (m)	0.0679	0.0651	0.0646	0.2734	0.0873	0.0886	0.0647	**0.0645**
	y (m)	0.0638	0.0629	0.0620	0.2225	0.0683	0.0828	0.0618	**0.0617**
	z (m)	0.0670	0.0659	0.0656	0.1331	0.0857	0.0809	0.0656	**0.0655**
	$\dot{x}$ (m/s)	0.3798	0.3689	0.3706	1.4747	0.5343	0.4936	0.3704	**0.3685**
	$\dot{y}$ (m/s)	0.3157	0.3059	0.3013	1.4103	0.4158	0.3976	0.3021	**0.3007**
	$\dot{z}$ (m/s)	0.2175	0.2080	0.2078	0.3212	0.2571	0.2365	0.2075	**0.2067**
	ω (rad/s)	0.1617	0.1654	0.1618	0.5816	0.1961	0.1875	0.1617	**0.1613**
8 observers	x (m)	0.0291	0.0285	0.0281	0.6575	0.7093	0.0331	0.0320	**0.0264**
	y (m)	0.1875	0.1804	0.1770	1.8786	2.6332	0.1994	0.2451	**0.1554**
	z (m)	0.0297	0.0295	0.0291	0.3601	0.2791	0.0330	0.0219	**0.0231**
	$\dot{x}$ (m/s)	0.1793	0.1717	0.1977	2.1985	2.2387	0.1859	0.1779	**0.1459**
	$\dot{y}$ (m/s)	0.0285	0.0284	0.0290	0.0844	0.0800	0.0295	0.0282	**0.0265**
	$\dot{z}$ (m/s)	0.0964	0.0987	0.1120	0.2117	0.3077	0.0997	0.1002	**0.0920**
	ω (rad/s)	0.0889	0.0836	0.0903	0.4140	0.3980	0.0964	0.1441	**0.0786**

- **Error-minimized ET filter module :** To assess the effectiveness of this module, we compare the performance of the EDC-CIF algorithm with and without the error-minimized ET filter. The former is a method without the incorporation of the proposed optimal event-triggered (ET) filter in the information form for diffusion fusion. As shown in Table I, our method achieves superior tracking accuracy and lower computation time compared to the one without the error-minimized ET filter under the same trigger rate, demonstrating the effectiveness of the proposed module in reducing estimation error and faster convergence speed.
- **Correlation-aware diffusion fusion module:** For this part of evaluation, we compare the EDC-CIF algorithm with and without the correlation-aware diffusion fusion, where the former only adopts a consensus-based fusion method without covariance intersection. As shown in Table I, compared to the version without correlation-aware diffusion fusion, the proposed EDC-CIF achieves lower estimation errors and faster computation under the same trigger rate, verifying the effectiveness of the proposed strategy in improving tracking accuracy without additional communication cost.

Note that the identical trigger rates of the algorithms arise from the use of the same ET mechanism, which transmits data based solely on differences between consecutive measurements. To ensure a fair comparison, both methods were tested on the same measurement series, resulting in consistent trigger rates.

APPENDIX G
MORE ALTERNATIVE COMPARISON

To ensure a fair and comprehensive comparison, we introduce an additional communication-efficient method in our experiments, DEC-CKF (Dynamic Event-Triggered Consensus-based Cubature Kalman Filter), which adopts a dynamic ET mechanism for collaborative tracking. This inclusion enables a thorough evaluation of both communication efficiency and tracking performance across different approaches for saving communication resources. As shown in Table I, our proposed EDC-CIF method outperforms DEC-CKF in terms of both position and velocity estimation accuracy, while also reducing the computation time. Moreover, this improvement is achieved under a relatively high trigger rate, further demonstrating the efficiency of our event-triggered design in balancing communication overhead and estimation accuracy.

APPENDIX H
COMPLETE EVALUATION RESULTS

A. Estimation Accuracy

As shown in Tab. II, most estimation errors decrease as the number of observers increases, demonstrating the effectiveness of scalable collaborative tracking in enhancing estimation accuracy. While non-ET schemes achieve small errors due to continuous communication (without reducing communication), the results clearly show that our proposed EDC-CIF outperforms all communication-efficient filter baselines in estimation accuracy for every state variable. Moreover, EDC-CIF consistently reduces estimation errors, even as the observer count increases. For example, EDC-CIF reduces x position estimation error by over 20% compared to the representative communication-efficient EC-CKF, as the observer count increases from 3 to 4 to 8. This highlights EDC-CIF's superior balance between communication efficiency and tracking accuracy.

B. Computation Time

The computation time results in Tab. III show that our proposed EDC-CIF consistently outperforms all baselines, even with varying observer counts. Specifically, EDC-CIF's

TABLE III
COMPUTATION TIME WITH DIFFERENT OBSERVER NUMBERS FOR ALL BASELINES.

Computation Time	**Non-ET schemes**			**ET schemes** (communication-efficient filters)				
	C-CKF	D-CIF	DC-CIF	EC-EKF	EC-UKF	EC-CKF	ED-CIF	EDC-CIF
3 observers t_{com} (ms)	1.28	0.97	0.99	1.20	1.39	1.38	0.98	**0.96**
4 observers t_{com} (ms)	2.37	1.33	1.54	2.22	2.24	2.50	1.37	**1.32**
8 observers t_{com} (ms)	4.69	2.09	2.56	2.97	3.77	4.94	2.19	**1.85**

advantage in reducing computation time becomes more pronounced as the number of observers increases. For example, compared to EC-CKF, EDC-CIF's computation time decreases by 30.43% (from 1.38 to 0.96 ms), 47.20% (from 2.50 to 1.32 ms), and 62.55% (from 4.94 to 1.85 ms) as the observer count increases from 3 to 4 to 8. Compared with non-ET schemes, EDC-CIF scales more efficiently as the number of observers increases. Compared with ET-based filters, EDC-CIF consistently incurs the least computational overhead. These results highlight the superior scalability and efficiency of EDC-CIF, making it a computationally efficient method for large-scale state estimation tasks.

APPENDIX I
ESTIMATION ACCURACY FOR REAL-WORLD EXPERIMENT

As for the real-world evaluation of our method's quantitative estimation accuracy, the RMSE results in Table IV show that the proposed EDC-CIF achieves the minimum estimation errors across all position state variables in the real-world UAV collaborative target tracking experiment. It consistently outperforms advanced EC-based filters and ED-CIF, demonstrating its effectiveness in reducing communication while maintaining precise state estimation.

TABLE IV
RMSE COMPARISON OF THE STATE VARIABLE IN THE REAL-WORLD MANUALLY CONTROLLED UAV EXPERIMENT.

State	EC-EKF	EC-UKF	EC-CKF	ED-CIF	EDC-CIF
x (mm)	9.2562	9.2261	9.1974	8.9307	**8.5531**
y (mm)	6.4443	6.3112	6.2543	6.2447	**6.2434**
z (mm)	6.5458	6.5326	6.5183	6.4419	**6.4414**

REFERENCES

[1] I. Arasaratnam and S. Haykin, "Cubature Kalman filters," *IEEE Transactions on Automatic Control*, vol. 54, no. 6, pp. 1254–1269, 2009.

[2] D. J. Tylavsky and G. R. Sohie, "Generalization of the matrix inversion lemma," *Proceedings of the IEEE*, vol. 74, no. 7, pp. 1050–1052, 1986.

[3] J. Hu, Z. Wang, H. Gao, and L. K. Stergioulas, "Extended Kalman filtering with stochastic nonlinearities and multiple missing measurements," *Automatica*, vol. 48, no. 9, pp. 2007–2015, 2012.

[4] W. Li, G. Wei, F. Han, and Y. Liu, "Weighted average consensus-based unscented Kalman filtering," *IEEE Transactions on Cybernetics*, vol. 46, no. 2, pp. 558–567, 2015.

[5] K. Xiong, H. Zhang, and C. Chan, "Performance evaluation of ukf-based nonlinear filtering," *Automatica*, vol. 42, no. 2, pp. 261–270, 2006.

[6] W. Li, G. Wei, F. Han, and Y. Liu, "Weighted average consensus-based unscented kalman filtering," *IEEE transactions on cybernetics*, vol. 46, no. 2, pp. 558–567, 2015.

[7] K. Reif, S. Gunther, E. Yaz, and R. Unbehauen, "Stochastic stability of the discrete-time extended Kalman filter," *IEEE Transactions on Automatic Control*, vol. 44, no. 4, pp. 714–728, 1999.

[8] K. Xiong, H. Zhang, and C. Chan, "Performance evaluation of UKF-based nonlinear filtering," *Automatica*, vol. 42, no. 2, pp. 261–270, 2006.

[9] G. Battistelli and L. Chisci, "Kullback–Leibler average, consensus on probability densities, and distributed state estimation with guaranteed stability," *Automatica*, vol. 50, no. 3, pp. 707–718, 2014.

[10] I. Matei and J. S. Baras, "Consensus-based linear distributed filtering," *Automatica*, vol. 48, no. 8, pp. 1776–1782, 2012.

[11] W. Song, J. Wang, S. Zhao, and J. Shan, "Event-triggered cooperative unscented Kalman filtering and its application in multi-UAV systems," *Automatica*, vol. 105, pp. 264–273, 2019.

[12] L. Li, D. Yu, Y. Xia, and H. Yang, "Event-triggered UKF for nonlinear dynamic systems with packet dropout," *International Journal of Robust and Nonlinear Control*, vol. 27, no. 18, pp. 4208–4226, 2017.